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# WHAT TYPES OF AI APPLICATIONS IN EDUCATION

Among the possible classifications of AI scientific domains, the following table presents an AI taxonomy[1] according to the family functions possibly performed by AI.

| | AI taxonomy | |
|---|---|---|
| | **AI domain** | **AI subdomain** |
| **Core** | **Reasoning** | Knowledge representation |
| | | Automated reasoning |
| | | Common sense reasoning |
| | **Planning** | Planning and Scheduling |
| | | Searching |
| | | Optimisation |
| | **Learning** | Machine learning |
| | **Communication** | Natural language processing |
| | **Perception** | Computer vision |
| | | Audio processing |
| **Transversal** | **Integration and Interaction** | Multi-agent systems |
| | | Robotics and Automation |
| | | Connected and Automated vehicles |
| | **Services** | AI Services |
| | **Ethics and Philosophy** | AI Ethics |
| | | Philosophy of AI |

*Figure: AI taxonomy - AI scientific domains and subdomains (from Samoili & al., 2021 JRC report[1]).*

Let's see which AI techniques are used in the AI-based education-oriented applications proposed by Holmes & al. in 2019[2].

| Student teaching | Student supporting | Teacher supporting | System supporting |
|---|---|---|---|
| Intelligent Tutoring Systems (ITS) (inc. automatic question generators) | Exploratory learning environments | ITS & learning diagnostics | Educational data mining for resource allocation |
| | Formative writing evaluation | Summative writing evaluation, essay scoring | |
| | Learning network orchestrators | Student forum monitoring | |
| Dialogue-based tutoring systems | Language learning applications | AI teaching assistants | Diagnosing learning difficulties |
| | AI collaborative learning | Automatic test generation | |
| | AI continuous assessment | Automatic test scoring | Synthetic teachers |
| | AI learning companions | Open Education Resources (OER) content recommendation | |
| Language learning applications (inc. pronunciation detection) | Course recommendation | Plagiarism detection | AI as a learning research tool |
| | Self-reflection support (learning analytics, meta-cognitive dashboards) | Student attention and emotion detection | |
| | Learning by teaching chatbots | | |

*Figure: Different types of current AI-based systems for Education (from Holmes & al. 2019[2]).*

Each AI-based educational tool or resource has its own specific techniques. However, it is sometimes possible to guess which ones are likely to be used for a given resource.

Let's take some examples:

- **Dialogue-based tutoring system**, as a student teaching service Such systems are likely to use: **communication** techniques such as natural language processing for speech and language understanding and generation and **reasoning** techniques for tutoring purposes
- **Course recommendation**, as a student supporting service As for personalised marketing offers and recommendations functions that can be found on the Internet, course recommendation systems are probably based on **machine learning** techniques by analysing relevant current data related to the student learning path and identifying similarities to previous generalised student learning paths.
- **Student attention and emotion detection**, as a teacher supporting service Such a system is likely to use **perception** techniques (computer vision for facial recognition for example) and **machine learning** techniques to analyse the student's facial expressions or behaviour if such information are collected and analysed.

1. AI Watch - Defining Artificial Intelligence - 2.0. Towards an operational definition and taxonomy for the AI landscape - Samoili, S., López Cobo, M., Delipetrev, B., Martínez-Plumed, F., Gómez, E., and De Prato, G. - EUR 30873 EN, Publications Office of the European Union, Luxembourg, 2021, ISBN 978-92-76-42648-6, doi:10.2760/019901, JRC126426. ↩↩
2. Artificial Intelligence In Education: Promises and Implications for Teaching and Learning - Wayne Holmes, Maya Bialik, Charles Fadel - Boston, MA, Center for Curriculum Redesign, 2019. ↩↩

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